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Research report

Human frontal eye fields and target switching

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ARTICLE INFO

Article history:
Received 27 August 2008
Reviewed 3 December 2008
Revised 31 December 2008
Accepted 23 January 2009
Action editor Elena Rusconi
Published online 3 March 2009

Keywords:
Frontal eye field
Transcranial magnetic stimulation
Task switching
Priming

ABSTRACT

The frontal eye fields (FEF) have typically been predominantly investigated in terms of their role in the generation of eye movements. Lesions to this area, either accidental or experimental, disrupt saccades and electrical stimulation elicits eye movements. Recently there has been increasing interest in the involvement of this area in visual processes, including in tasks where eye movements were either not required or were precluded. In addition to being involved in a range of visual tasks, evidence from visual search paradigms has suggested that this area might be important when the defining quality of the target is unpredictable or that it may be involved in priming. We investigated the role of FEF in a task requiring localisation of a target defined by colour, in which the target colour was either maintained or switched across trials. Disruption of performance was seen on the task when transcranial magnetic stimulation (TMS) was delivered over the left FEF, specifically elevating response times on trials when the target and distracter colours were switched rather than affecting any benefit of repetition of the target attribute (priming). This result is consistent with altered modulation of extrastriate areas, consequently affecting the speed with which a switch of the target colour could be detected. This both offers an explanation for effects seen in unpredictable feature search and is consistent with other TMS and microstimulation studies showing that FEF modulates responses of extrastriate cortex.

1. Introduction

The frontal eye field is a region of brain lying anterior to the motor cortex that is typically associated with occulomotor control. Lesions to this region in monkeys produce disruptions to saccade generation (Latto and Cowey, 1971; Rizzolatti, 1983). In humans, electrical stimulation evokes saccades (Godoy et al., 1990; Rasmussen and Penfield, 1948) as well as in monkeys (Russo and Bruce, 1993). Transcranial magnetic stimulation (TMS) to the region interferes with saccades (Muri et al., 1991; Thickbroom et al., 1996). While this is, at least until relatively recently, the function with which this area has been predominantly associated (e.g., Schiller and Chou, 1998; Tehovnik et al., 2000), a role for this area in visual processing

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0010-9452/$ – see front matter © 2009 Published by Elsevier Srl.
doi:10.1016/j.cortex.2009.01.011

independent from a role in eye movements has previously been suggested (Latto and Cowey, 1971). Furthermore, the role of the frontal eye fields (FEF) in visual processes has been the subject of electrophysiological studies in macaques (e.g., Schall, 1997; Bichot and Schall, 1999), where responses consistent with visual selection by FEF neurons have been reported. In humans, transcranial magnetic stimulation studies have shown the necessity of FEF in visual tasks in the absence of eye movements (Muggleton et al., 2003; O'Shea et al., 2004). Furthermore, microstimulation, and subsequently TMS, has provided evidence for a dissociation of saccade programming and visual processing in this area (Juan et al., 2004, 2008).

Previously we have found that, like posterior parietal cortex (Ashbridge et al., 1997; Walsh et al., 1998), TMS delivered over FEF disrupted performance on visual search for a target defined by a combination of attributes (shape and colour, conjunction search), but not by a single feature (colour) (Muggleton et al., 2003; O'Shea et al., 2004; Kalla et al., 2008). This dissociation was clear for conjunction search versus feature search when the target colour in feature search was consistent across trials (e.g., the target to be detected was always red with distracters always blue). However, in the case of a feature search task when the target colour could switch randomly across trials, the case was less clear (Muggleton et al., 2003). For this task, the effects of TMS were intermediate between the disruption seen for conjunction search and the absence of any effect on feature search with a consistent target colour. This raised the question of whether FEF is important when an unpredictable target must be detected. There are two ways in which performance might be affected. First, TMS over FEF might disrupt any beneficial effects of repeating the target attribute across trials. This priming of popout effect is the well-known benefit (in terms of either speed or accuracy) which results from repetition of attributes across trials, compared with trials when such attributes change (Maljkovic and Nakayama, 1994, 1996, 2000). In the case of the unpredictable feature search task, this would reflect disruption of colour priming (Maljkovic and Nakayama, 1994). Priming effects may also occur for position (Maljkovic and Nakayama, 1996) but the target location in the search task was random, so this was not a factor (present/absent judgments could also be primed, but this was the case for all three of the search tasks). A second possible explanation for the modulation of performance seen could be that the ability to detect that the target colour and distracter colour had switched was affected by TMS delivered over FEF.

It has been suggested that the brain region responsible for the encoding of an attribute is involved in short-term memory for that type of attribute (Magnussen, 2000; Tulving and Schacter, 1990). Lesion (Walsh et al., 2000; Bisley and Pasternak, 2000), imaging (Henson, 2003) and psychophysics evidence (Bar and Biederman, 1999; Maljkovic and Nakayama, 1994, 1996, 2000) has been found to support this theoretical position, as have TMS studies (Campana et al., 2002, 2006). The role of FEF in priming has also been recently investigated using TMS (Campana et al., 2007), employing a TMS paradigm previously found to be effective in providing evidence for the role of area MT/V5 in motion direction priming but not spatial priming of motion (Campana et al., 2002). Their results suggested a role for left (but not right) frontal eye field in priming for spatial position but not for stimulus attributes. These data show that FEF plays a role in priming, but does not offer an explanation for the effects seen on unpredictable feature search, as spatial priming of target position did not occur in the search task. Furthermore, TMS delivered over FEF in a feature priming task where responses were saccades to the target showed effects on colour priming, but only when TMS was delivered during stimulus presentation (O'Shea et al., 2007). FEF neurons are not themselves colour selective (Mohler et al., 1973), so would not be expected to mediate colour priming according to the idea that areas representing an attribute are involved in memory for that attribute (Tulving and Schacter, 1990). However, colour related modulation of FEF activity has been recorded from FEF neurons (Bichot and Schall, 2002), with earlier target selection in visuo-movement neurons and a greater target-distractor activity difference following selection. In imaging studies, feature (or location) repetition results in suppression of responses in FEF (Kristjansson et al., 2007), an effect that correlates with priming (Desimone, 1996). Microstimulation of FEF in non-human primates has been shown to modulate visual responses to targets in area V4 (Moore and Fallah, 2001; Moore and Armstrong, 2003) and lesions to V4 have been found to abolish colour priming (Girard et al., 2001; Rossi et al., 2001; Walsh et al., 2000). Finally, we have recently shown that TMS to FEF changes the sensitivity of FEF within a short time window (Silvanto et al., 2006). It is possible, then, that effects of FEF TMS on detection of an unpredictable coloured target might therefore be a consequence of altered modulation of extrastriate cortex by FEF.

The aims of the experiment reported here were to (1) establish more clearly whether or not FEF is involved in a task where the target defining attribute (colour) was not fixed across trials and (2) to evaluate whether any disruption (if seen) was a consequence of disruption of priming (or modulation of the effects of switching of the target attribute), in the absence of eye movements and with no spatial priming. We therefore presented a target localisation task, where the target was defined as the odd coloured element, using small displays such that eye movements were unnecessary for performance and investigated the effects of TMS over right and left FEF. While the target colour could either be maintained or switched across trials, locations were never repeated to eliminate possible effects on spatial priming.

2. Methods

2.1. Subjects

Seven neurologically intact right-handed participants (5 males, 2 females) aged 22–41 years took part in the experiment. All gave informed consent prior to participation and the experiment was approved by the Oxford Research Ethics Committee (OxREC) and the Institute of Neurology, University College, London. Participants were naïve to the purpose of the experiment and also to the task prior to taking part.

2.2. TMS and site localisation

Transcranial magnetic stimulation was delivered by means of a Magstim Super Rapid TMS machine (Magstim Company, Dyfed, UK). A 50 mm figure of eight coil was used to deliver 5 pulses at a frequency of 10 Hz during the task over either right FEF, left FEF or right posterior parietal cortex (PPC). A fixed level of 60% of the maximum machine output was employed. Stimulation was not related to motor threshold as this has been shown to not be a good guide to effective stimulation levels in other brain regions (Stewart et al., 2001). The level selected was used as it has previously proved effective in TMS studies (e.g., Kalla et al., 2008). Stimulation of FEF at levels higher than this has not been associated with disruption that could be attributed to non-specific stimulation (e.g., Juan et al., 2008; O'Shea et al., 2007; Muggleton et al., 2003). Sites were localised for each subject using a high resolution ($1 \times 1 \times 1$ mm) structural magnetic resonance image (MRI) scan. Each scan was normalised individually against a standard template using FSL (FMRIB, Oxford) which produced both a normalised scan as well as details of the normalisation transformation for that individual. This transformation was then reverse applied to the coordinates of the sites of interest. These were −32, −2, 46 (left FEF, Paus, 1996), 31, −2, 47 (right FEF, Paus, 1996). Effective stimulation of each of these has been previously shown in a range of studies (Göbel et al., 2001; Muggleton et al., 2003; O'Shea et al., 2004). Visual inspection confirmed that for all subjects the transformed coordinates resulted in acceptable localisation of the FEF with respect to the typical anatomical description of its location. The vertex site was localised as the point lying midway between the inter-tragal notches of the ears and midway between the inion and nasion.

2.3. Task

The experimental timeline is shown in Fig. 1. Following a 500 msec fixation the stimulus was presented. This consisted of 4 small squares (.2 × .2 degrees) placed on the corners of an imaginary 2 degree square. Three of these were the same colour (either red or blue, isoluminant, 20 cd.m^3) and the final one was the other colour and acted as the target (i.e., a red target with blue distractors or *vice versa*). A response was required, as quickly and accurately as possible, indicating the position in which the odd coloured element had appeared. This was achieved by means of the 1,2,4 and 5 buttons of the number pad of a standard keyboard, with their square arrangement corresponding to the locations in which target could appear. Subjects were instructed to press the appropriate key with the index finger of the right hand. Following a response the screen went blank for 500 msec, TMS was then delivered at 10 Hz for 500 msec and followed by a further 500 msec blank before the next trial began. These stimulation timings were selected on the basis of previous studies by Campana et al. (2002, 2006, 2007), where inter-trial TMS was employed to investigate the contribution of different areas to priming effects. The target location did not repeat from one trial to the next and three trials were presented with the same target colour. After these three trials, there was a fifty percent chance of a colour switch on the next trial/trials (after a switch this process began again). This approach was used to try and ensure the cost of switching/benefit of priming represented a significant percentage of the response times and to facilitate detection of any reduction in priming effects that might possibly occur. Subjects were not informed of the rules governing when a switch may occur. Two blocks, each containing 40 trials, were presented per site for each subject and the order of presentation was randomised.

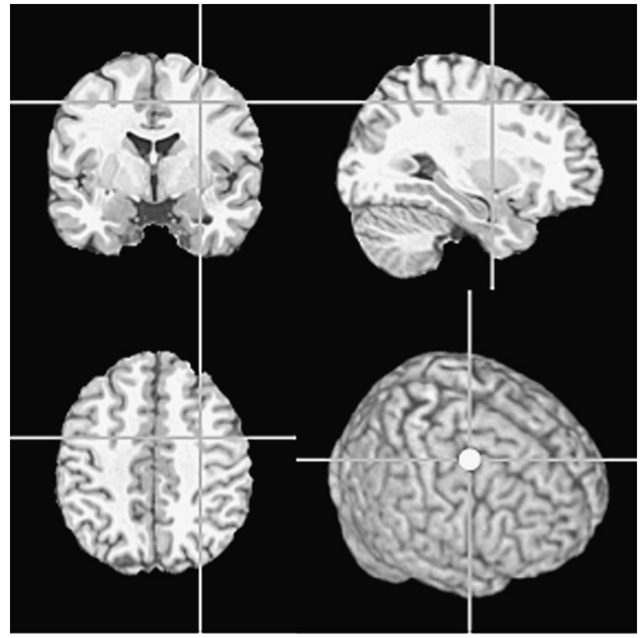

Fig. 1 – Localisation of the frontal eye field shown for one subject. Locations for stimulation were calculated using the FSL software package (FMRIB, Oxford, UK) and the points on the scalp overlying these identified using the Brainsight system. TMS was delivered over these points in the inter-trial interval.

3. Results

Data were analysed primarily in terms of response times to localise the target and are illustrated in Fig. 2. Few errors were made (mean accuracy = 95%), and were excluded from analysis when they occurred. Response times greater than two standard deviations from the mean were also excluded. Three types of trial were included in the analysis; trials where a switch occurred, the first repeat of a colour and the second repeat. There was the same number of each these trial types as three consecutive presentations of a colour were necessary before the possibility of a switch. Two analyses were conducted, each a repeated measures analysis of variance. Both used trial type as the independent variable and the first took the differences between the response times as the dependant variable, i.e., switch RT – first repeat RT (priming effect size) and first – second repeat RT (repeat priming effect). This was therefore an analysis of the effects of TMS on priming effects (see Fig. 3). The second took response time as the dependant

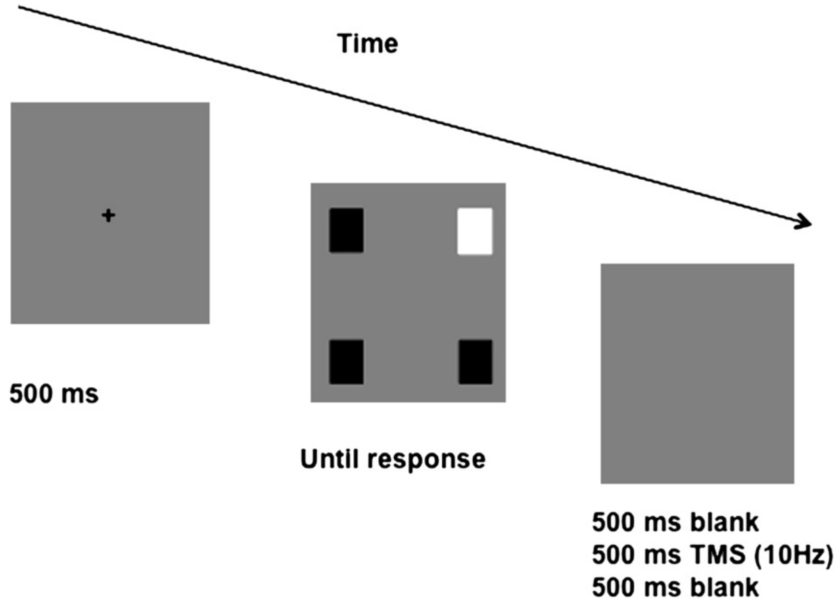

Fig. 2 – Timeline of the experimental procedure. Following fixation, the stimulus (in which the target squares were positioned such that they formed the corners of a 2 degree square) was presented until a response was made. During the subsequent 1500 msec inter-trial interval, TMS was delivered at 10 Hz for 500 msec in the middle of this period.

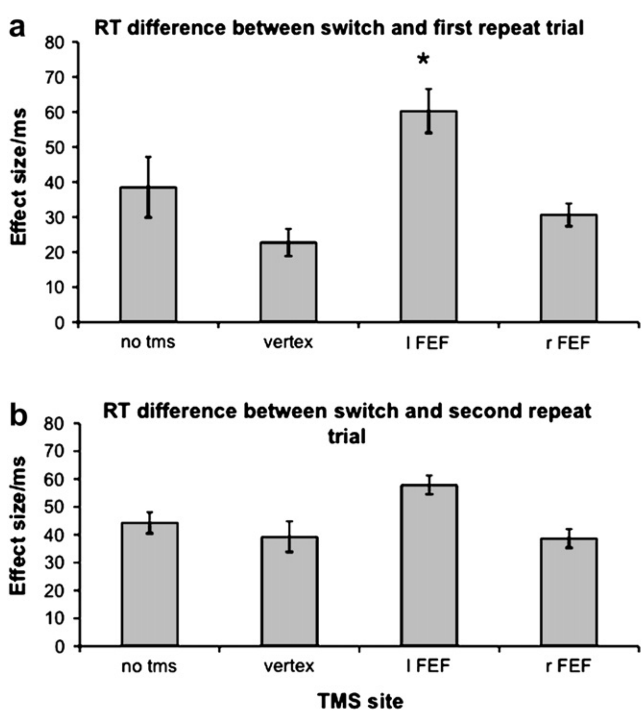

Fig. 3 – Effect of TMS on priming. (a) Response time differences (± standard error of the mean) between the trial on which the target colour was changed and the first repeat (larger values represent a greater improvement in response speed on repeat of a colour). (b) Response time differences between the trial on which the target colour was changed and the second repeat of the colour. Left frontal eye field TMS resulted in an elevated priming effect for the first repeat of the target colour following it having switched. Error bars represent the standard error of the mean.

variable and the independent variable of time had three levels (corresponding to switch trials, first primed trial and second primed trial) (see Fig. 4).

The first analysis showed a significant main effect of TMS site [F (3,18) = 5.48, $p = .007$], a main effect of prime type (first or second) [F (1,6) = 22.886, $p = .003$]. The TMS effect was due to greater priming with FEF TMS compared to all of the other sites (*vs* no TMS, mean difference = 6.8 msec, $p = .024$; *vs* vertex, mean difference = 9.3 msec, $p = .024$; *vs* right FEF, mean difference = 9.6 msec, $p = .003$). The time effect was due to greater priming on the first versus the second repeat of a stimulus (mean difference = 31.1 msec, $p = .003$). To investigate the interaction, priming with left FEF TMS was compared to the other sites for the first and second primes using paired *t*-tests (corrected for multiple comparisons). For

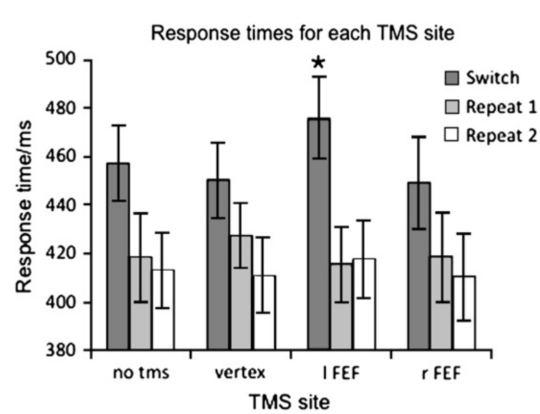

Fig. 4 – Effects of TMS on response times for trials when the target colour was switched and the subsequent two repeats of the target colour. Error bars have been omitted for clarity but ranged from a minimum of 13.7 to a maximum of 19.1 msec (standard error of the mean).

the first time interval, left FEF differed significantly from all other sites (no TMS, $t = 3.304$, $p = .039$; vertex, $t = 4.293$, $p = .012$; right FEF, $t = 3.878$, $p = .020$). For the second time interval, left FEF did not differ from the other conditions.

The second analysis (of response times) showed a main effect of time [F $(2,12) = 118.1$, $p < .001$] and a time × site interaction [F $(6,36) = 5.172$, $p = .001$]. The main effect of time was due to slower responses on the switch trials compared to the subsequent repeat trials (primed *vs* first repeat, mean difference = 38.1 msec, $p < .001$; primed *vs* second repeat, mean difference = 45.0 msec, $p < .001$). To investigate the interaction, the site for which disruption of priming was seen (left FEF) was compared to the other stimulation conditions for each time interval using paired t-tests (corrected for multiple comparisons). These revealed elevated response times on switch trials for left FEF stimulation versus both vertex ($t = 3.852$, $p = .024$) and right FEF ($t = 3.770$, $p = .027$).

In summary, the analysis of the data showed that left FEF TMS significantly elevated response times on the trial when the target colour had switched.

4. Conclusions

Our findings clearly show that the left FEF is important in selecting a new target – i.e., a target that was a distracter on a previous trial.

It has previously been shown that the FEF are involved in search for a target defined by a combination of attributes (colour and shape, conjunction search) (Muggleton et al., 2003; O'Shea et al., 2004; Kalla et al., 2008), but not when defined by a single feature (colour, feature or popout search) (Muggleton et al., 2003). Here we investigated the ambiguous effect seen when the target defining element in a feature search task was unpredictable across trials (Muggleton et al., 2003).

Transcranial magnetic simulation delivered over left, but not right, FEF disrupted performance on the feature target localisation task. TMS elevated the difference in response times between the first presentation of the target colour and the second occurrence of the colour (i.e., an apparent increase in the priming effect of repeating the target colour). However, rather than being a consequence of greater priming, this effect was due to elevated response times specifically for the trials when the target colour and distracter colours switched. Such an effect would be consistent with altered performance for an unpredictable feature search.

As well as offering an explanation for the pattern of performance previously seen on unpredictable feature search, the effect reported here raises several questions. Whereas the effect on feature search was a consequence of TMS delivered over right FEF, that seen here was a result of left FEF stimulation (a site not tested in the unpredictable feature search task). The evidence that left FEF is involved in unpredictable feature search presented here provides some insight into the hemispheric asymmetries of the FEF which have been less well explored than asymmetries in the parietal lobe. Regions in the left hemisphere such as the supplementary motor area (Rushworth et al., 2002; Kennerley et al., 2004) and intraparietal sulcus (Rushworth et al., 2001) are important for attention to action, action selection and action sequencing and it seems reasonable to suggest that the left FEF forms a component of this left hemisphere choice-action system. There are also different requirements of the two tasks that may have played a part in evoking different effect of TMS. While they both involve detection of a colour-defined singleton, the present study requires the localisation of a target that is present on every trial. In the unpredictable feature task (Muggleton et al., 2003) detection of the target, which was present in half of the trials, was required and no localisation was necessary and it has been argued that successful feature search does not require target localisation (e.g., Treisman, 1998). It may be this difference in task requirements which results in differences in the hemisphere critical for task performance.

The left FEF has previously been shown to be involved in priming. Campana et al. (2007), using a task where subjects were required to either localise the odd direction target or indicate its direction, showed that left FEF was involved in priming for spatial position but not for motion direction. It has been suggested that the FEF hold a saliency map with respect to the target (Schall and Bichot, 1998) making this finding consistent with the idea that areas involved in the processing of a stimulus' quality are involved in short-term memory for that attribute. Similarly, O'Shea et al. reported that TMS over left FEF during (but not between) trials abolished spatial priming of popout, in a task where the responses were eye movements to the target. It was suggested that left FEF might integrate a spatial memory signal when a saccade is made to a repeated location. In the present study no location priming was possible as the task was constructed in such a way to avoid this, as the target location in the unpredictable popout task was random across trials. It remains to be determined if the pattern of results seen in the present study is a consequence of the requirements for a manual, rather than saccade localisation of the target or if they are a consequence of the less frequent occurrence of switches in the target and distractor attributes.

One explanation for the effects seen here is that TMS over left FEF alters modulation of extrastriate areas with respect to the target and distracters, a role suggested by microstimulation studies in the macaque (Moore and Armstrong, 2003). It is possible that TMS induced modulation changes produce little effect on trials with repeats, failing to disrupt target detection/localisation, but impairs the ability to detect that a change has occurred on switch trials, thus elevating response times. Future work might more specifically address this issue by, for example, having unpredictable distracters, or evaluating separable motor and visual components of task performance, an approach successfully employed in saccade tasks (Carpenter and Williams, 1995).

The findings here extend the processes to which human FEF are known to contribute to in visual tasks and suggest that, in addition to being important for detection of a target defined by a combination of features (Muggleton et al., 2003) and having a role in spatial priming (Campana et al., 2007; O'Shea et al., 2007) as well visual awareness (Grosbras and Paus, 2003) and visual cuing (Smith et al., 2005), they are involved in detection of switching of target and distractor attributes.

Acknowledgements

This work was supported by the UK Medical Research Council. CHJ was supported by the National Science Council, Taiwan (96-2413-H-008-001, 97-2511-S-008-005-MY3).

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